

UNIT 2

Time Series Regression Model and
Exponential Smoothing

By

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Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

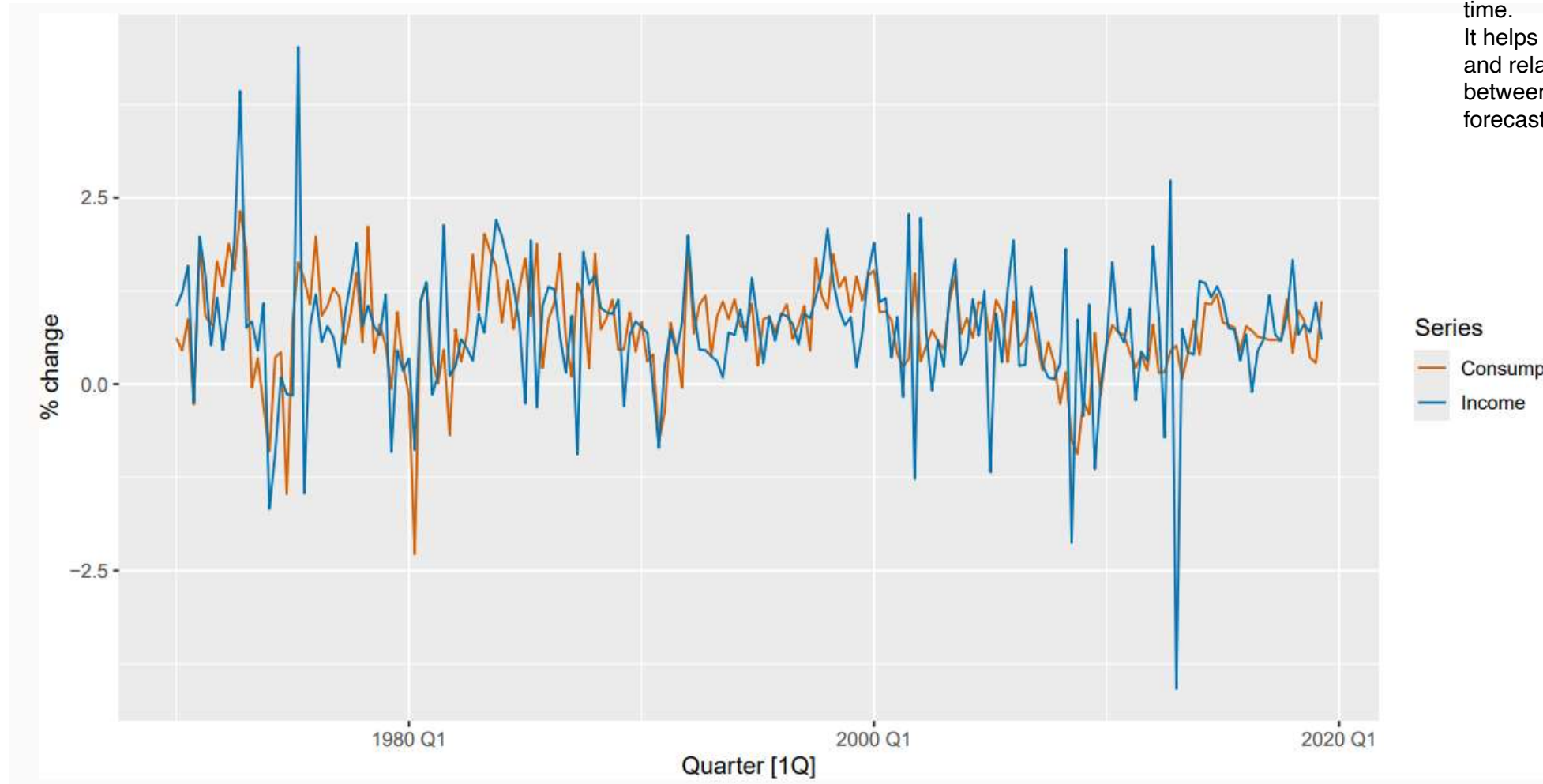
- y_t is the variable we want to predict: the “response” variable
- Each $x_{j,t}$ is numerical and is called a “predictor”. They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \dots, \beta_k$ measure the effect of each predictor after taking account of the effect of all other predictors in the model.

That is, the coefficients measure the **marginal effects**.

- ε_t is a white noise error term

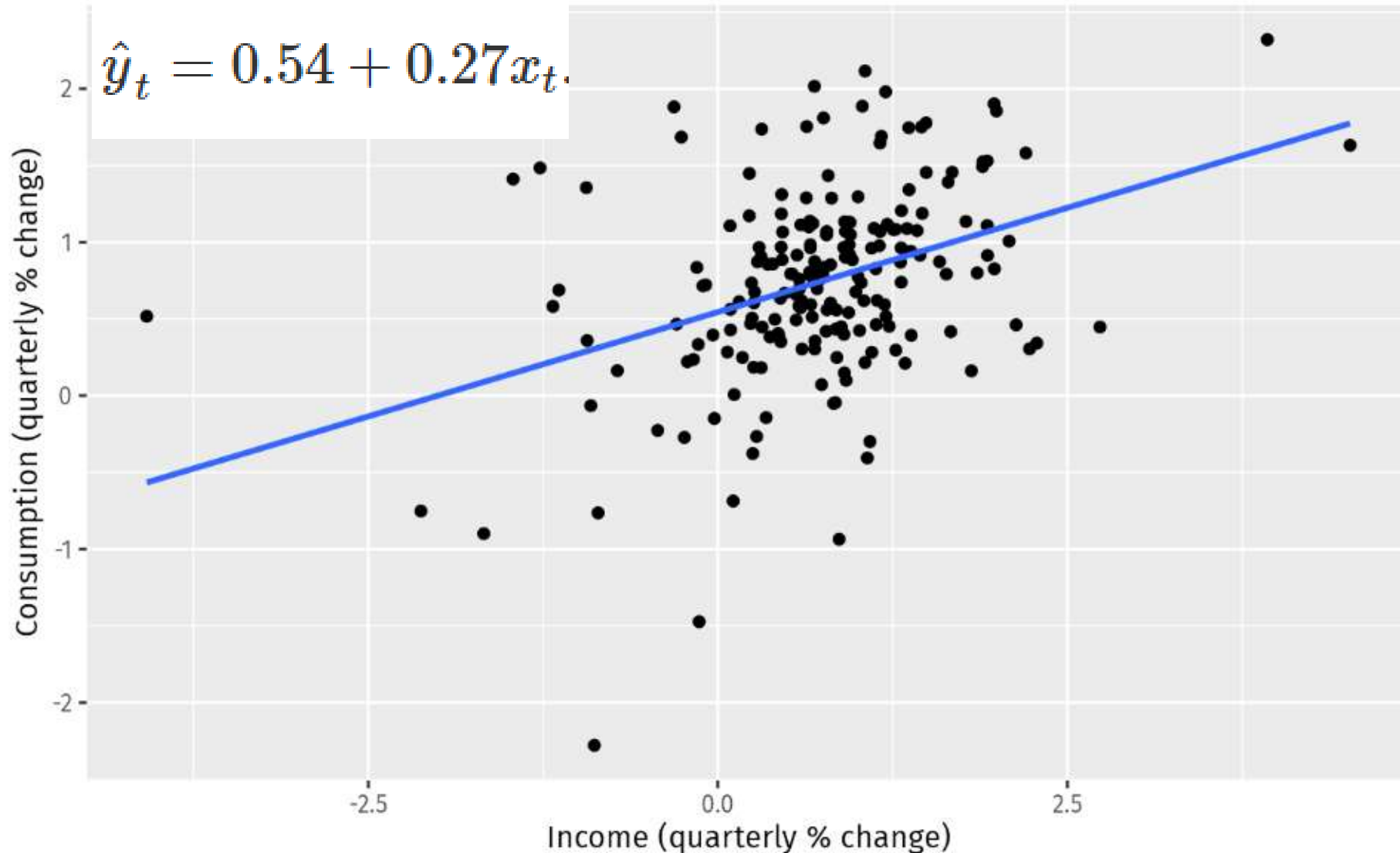
Example: US consumption expenditure

Shows how consumption and income change over time.
It helps identify trends and relationships between variables for forecasting.



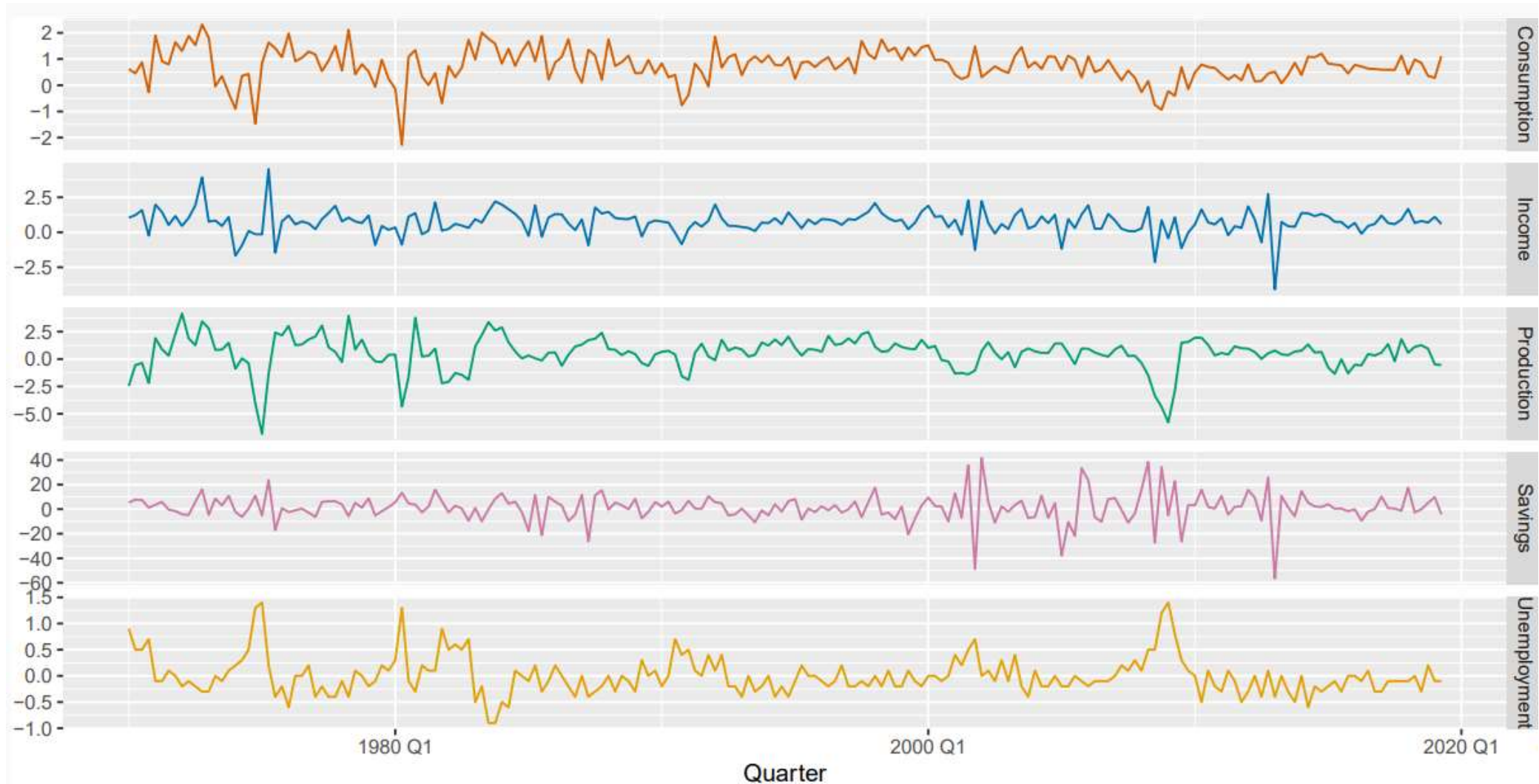
Example: US consumption expenditure

Shows the relationship between income (predictor) and consumption (output). The regression line represents the equation used to predict consumption from income.



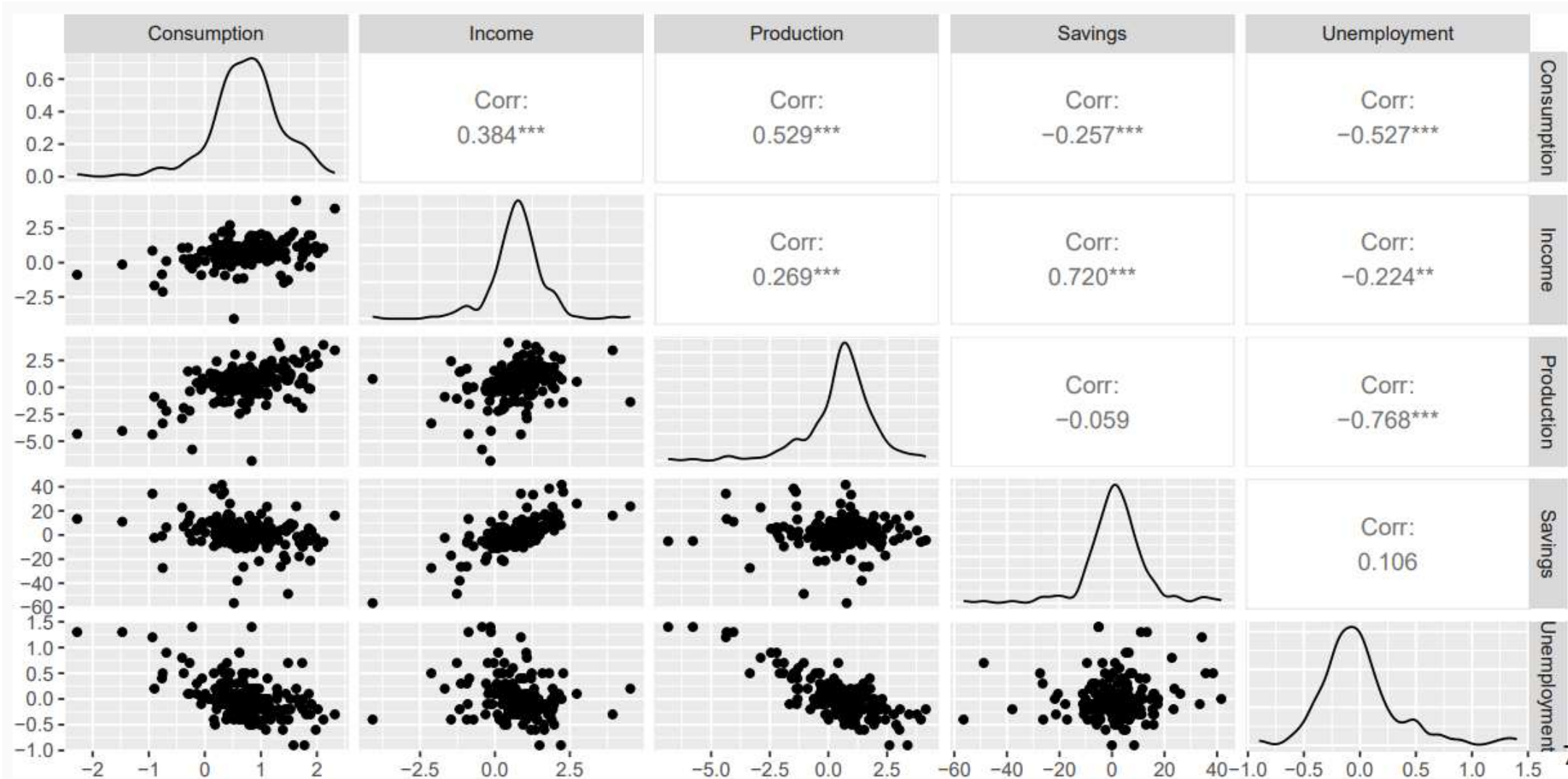
Example: US consumption expenditure

Shows multiple variables changing over time together. These variables are used as predictors to forecast consumption using multiple regression.



Example: US consumption expenditure

Shows the strength and direction of relationship between all variables. It helps identify which variables are useful predictors for regression.



Assumptions for the linear model

For forecasting purposes, we require the following assumptions:

- ε_t have mean zero and are uncorrelated.
- ε_t are uncorrelated with each $x_{j,t}$.

It is **useful** to also have $\varepsilon_t \sim N(0, \sigma^2)$ when producing prediction intervals or doing statistical tests.

Least squares estimation

- In practice we need to estimate the coefficients: $\beta_0, \beta_1, \dots, \beta_k$.

$$\sum_{t=1}^T \varepsilon_t^2 = \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t} - \dots - \beta_k x_{k,t})^2$$

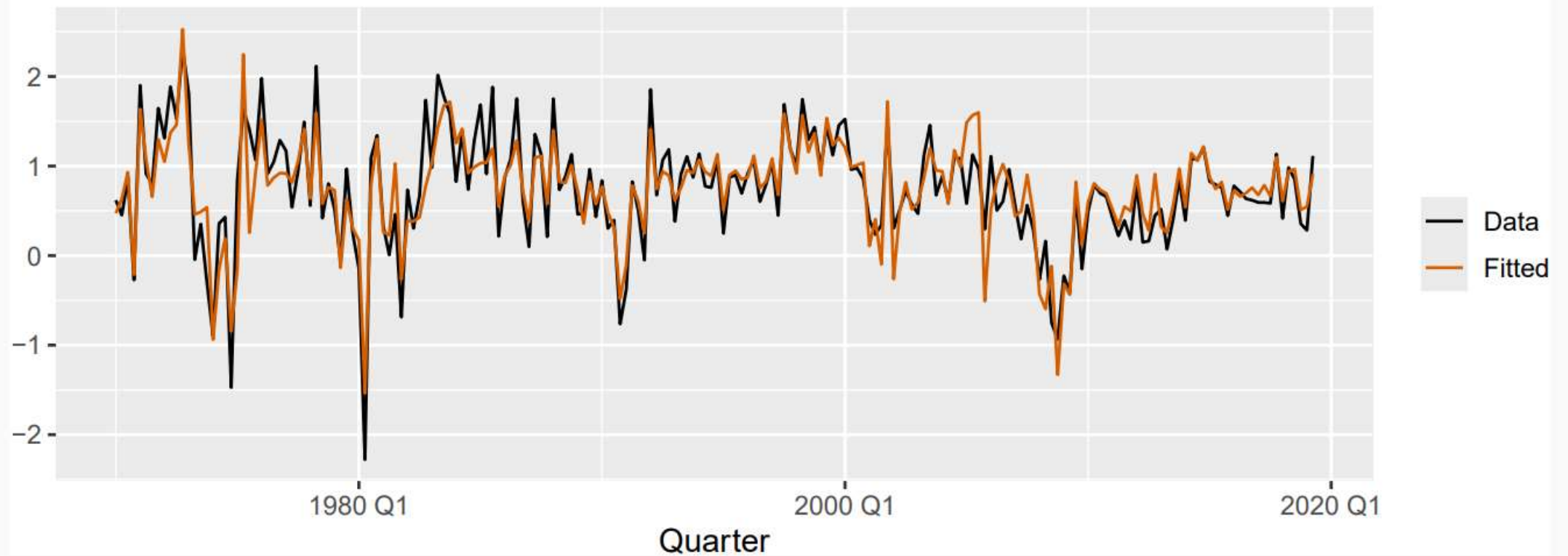
```
model(TSLM(y ~ x_1 + x_2 + ... + x_k))
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- Estimated coefficients: $\hat{\beta}_0, \dots, \hat{\beta}_k$

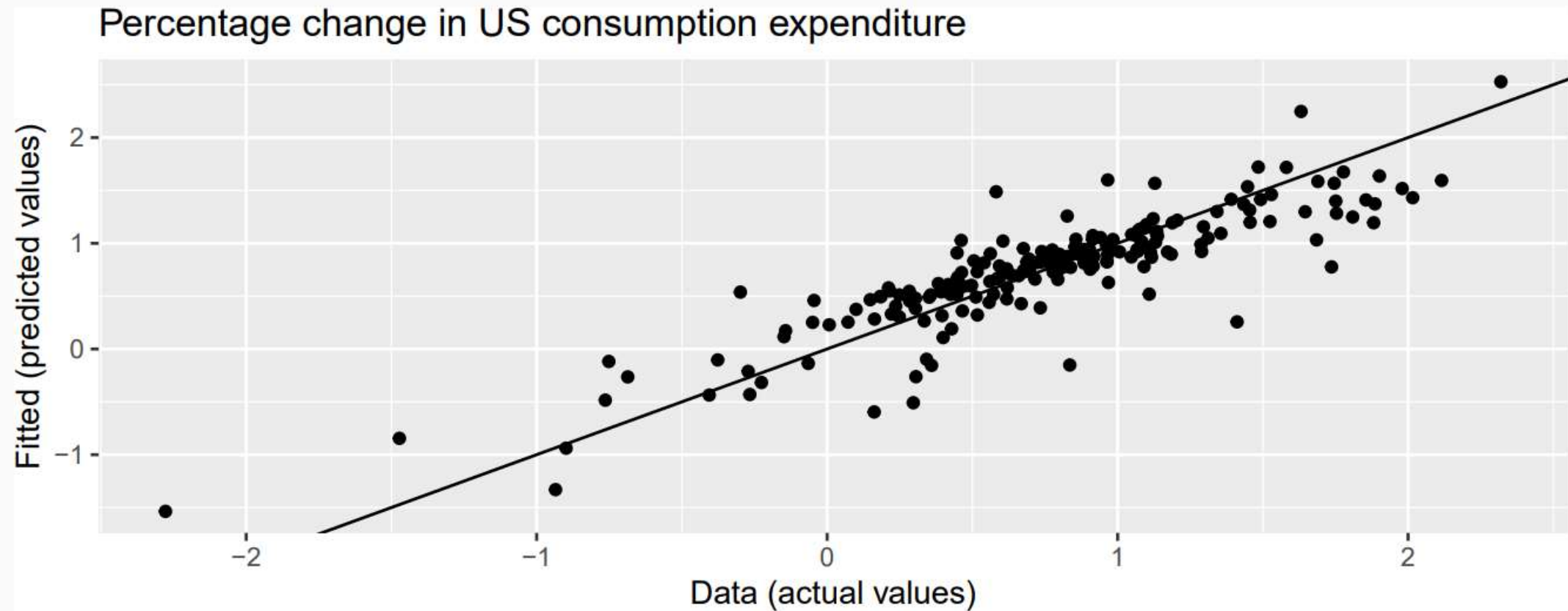
Fitted values

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1,t} + \hat{\beta}_2 x_{2,t} + \dots + \hat{\beta}_k x_{k,t}$$

Percent change in US consumption expenditure



Example: US consumption expenditure



Goodness of fit

Coefficient of determination

$$R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2}$$

Standard error of the regression

$$\hat{\sigma}_e = \sqrt{\frac{1}{T - k - 1} \sum_{t=1}^T e_t^2}$$

where k is the number of predictors in the model.

Evaluating the regression model

Regression
residuals

$$\begin{aligned} e_t &= y_t - \hat{y}_t \\ &= y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t} - \cdots - \hat{\beta}_k x_{k,t} \end{aligned}$$

Useful properties

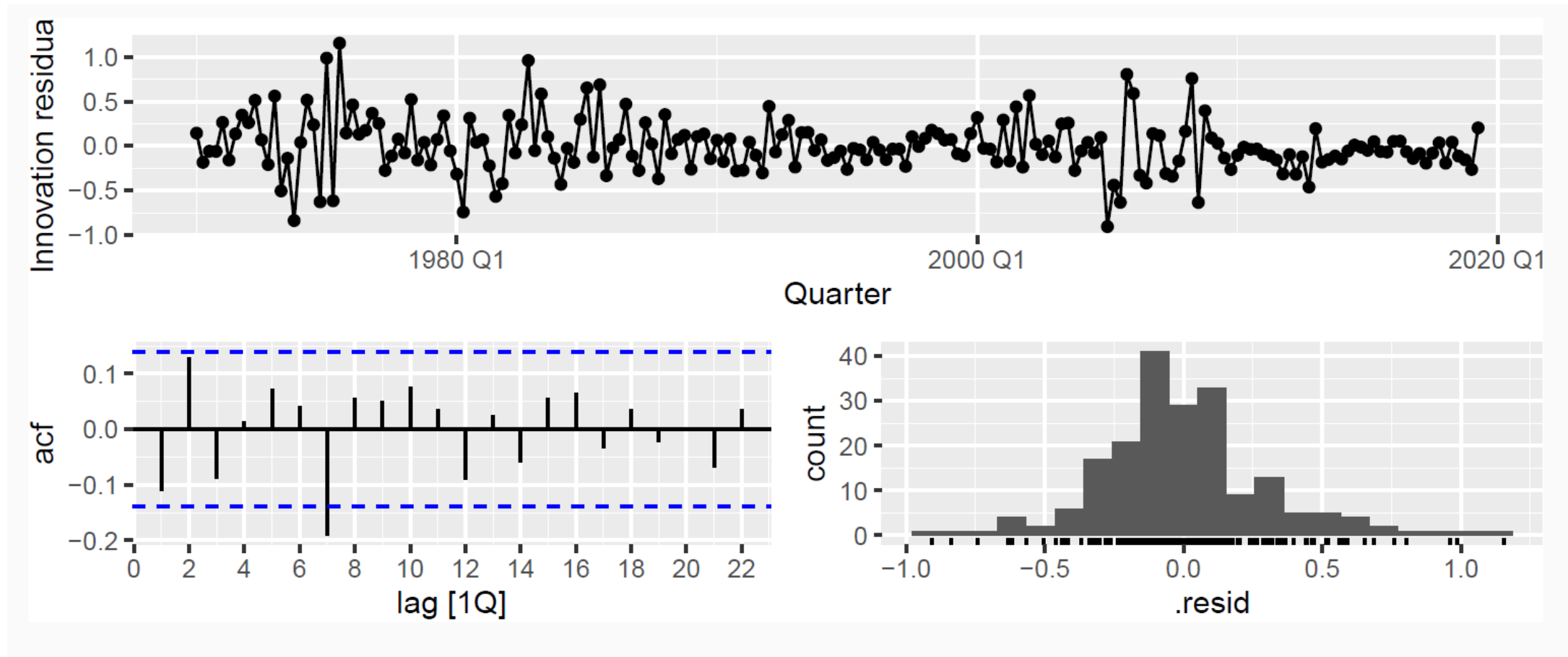
$$\sum_{t=1}^T e_t = 0 \quad \text{and} \quad \sum_{t=1}^T x_{k,t} e_t = 0 \quad \text{for all } k.$$

Checking assumptions

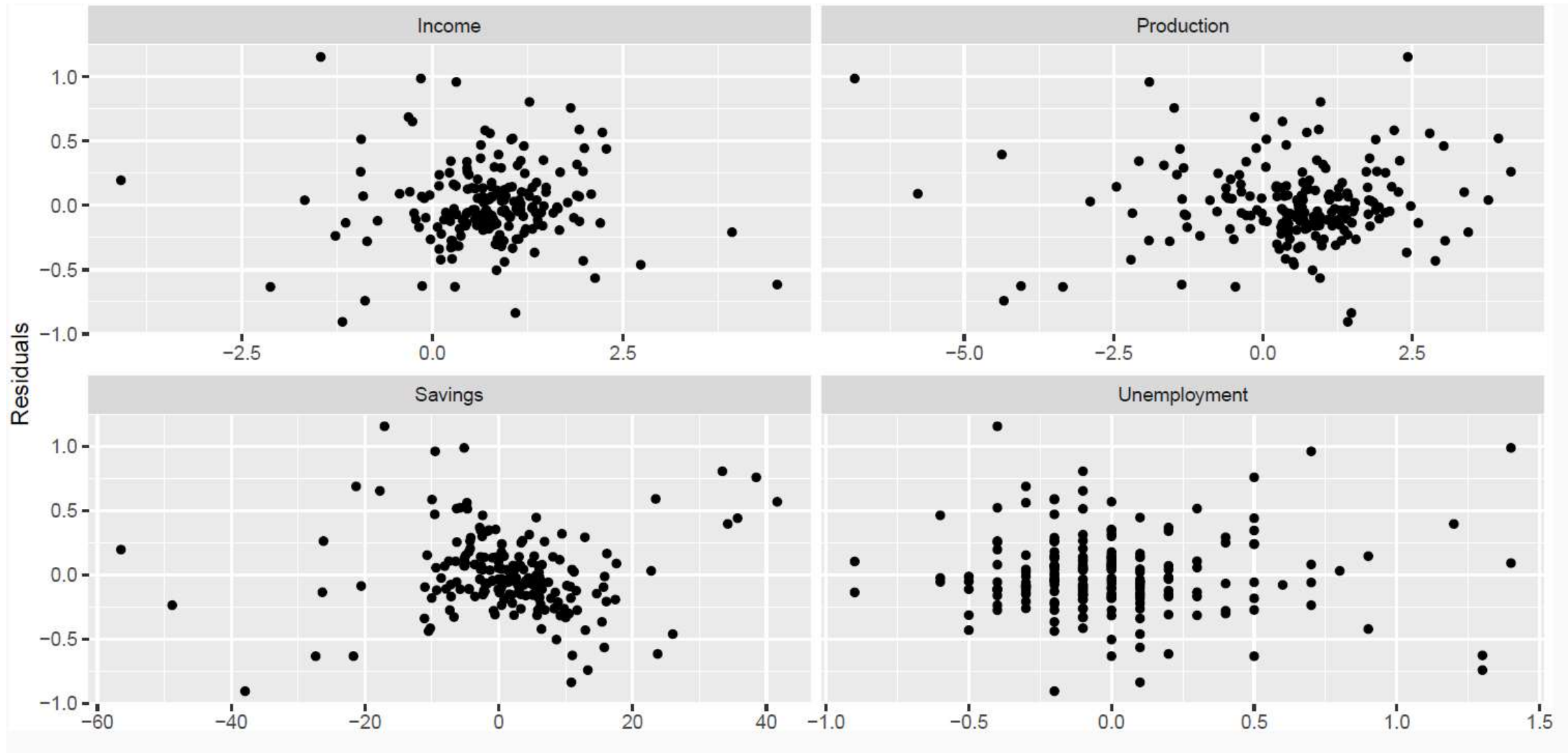
- ε_t have mean zero and are uncorrelated, $\text{NID}(0, \sigma^2)$.
- ε_t are uncorrelated with each $x_{j,t}$.
 - ▶ Timeplot, ACF, Histogram (`gg_tsresiduals()`)
 - ▶ Against predictors (non-linearity)
 - ▶ Against fitted values (heteroscedasticity)
 - ▶ Against predictors not in the model (include predictor in the model)

Expect to see scatterplots resembling a horizontal band with no values too far from the band and no patterns such as curvature or increasing spread.

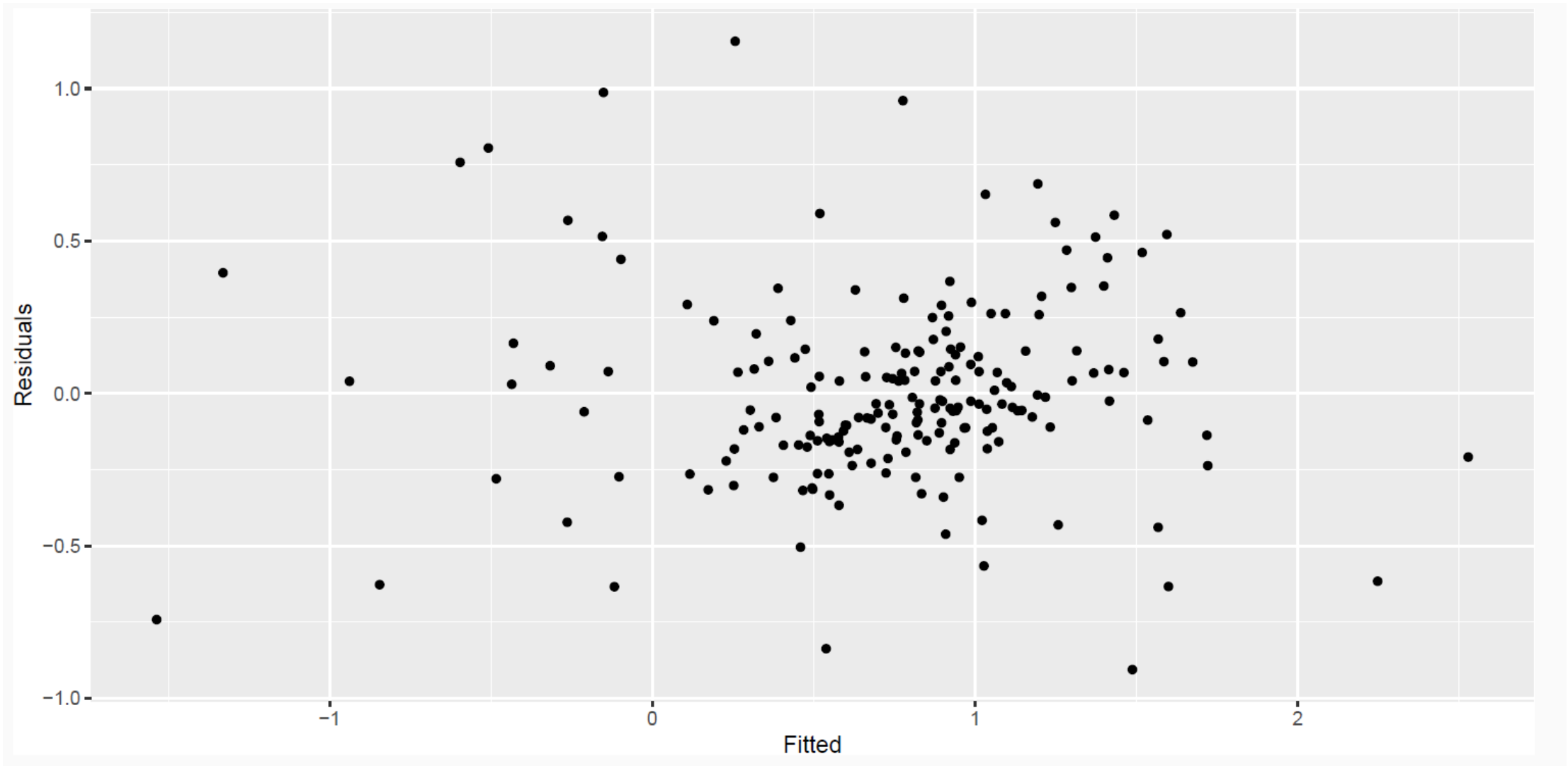
Example: US consumption expenditure



Example: US consumption expenditure



Example: US consumption expenditure



Selecting predictors

Computer output for regression will always give the R^2 value. This is a useful summary of the model.

- It is equal to the square of the correlation between y and $\hat{y}$.
- It is often called the “coefficient of determination”.
- It can also be calculated as follows:

$$R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2}$$

- It is the proportion of variance accounted for (explained) by the predictors.

Comparing regression models

However ...

- R^2 does not allow for “degrees of freedom”.
- Adding *any* variable tends to increase the value of R^2 , even if that variable is irrelevant.

To overcome this problem, we can use *adjusted* R^2 :

$$\bar{R}^2 = 1 - (1 - R^2) \frac{T - 1}{T - k - 1}$$

where k = no. predictors and T = no. observations.

Maximizing $\bar{R}^2$ is equivalent to minimizing $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T - k - 1} \sum_{t=1}^T \varepsilon_t^2$$

Akaike's Information Criterion

$$\text{AIC} = -2 \log(L) + 2(k + 2)$$

where L is the likelihood and k is the number of predictors in the model.

- AIC penalizes terms more heavily than $\bar{R}^2$.
- Minimizing the AIC is asymptotically equivalent to minimizing MSE via **leave-one-out cross-validation** (for any linear regression).

Corrected AIC

For small values of T , the AIC tends to select too many predictors, and so a bias-corrected version of the AIC has been developed.

$$\text{AIC}_C = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

As with the AIC, the AIC_C should be minimized.

Bayesian Information Criterion

$$\text{BIC} = -2 \log(L) + (k + 2) \log(T)$$

where L is the likelihood and k is the number of predictors in the model.

- BIC penalizes terms more heavily than AIC
- Also called SBIC and SC.
- Minimizing BIC is asymptotically equivalent to leave- v -out cross-validation when $v = T[1 - 1/(\log(T) - 1)]$.

Exponential smoothing

Simple methods

Time series $y_1, y_2, \dots, y_T$.

Random walk forecasts

$$\hat{y}_{T+h|T} = y_T$$

Average forecasts

$$\hat{y}_{T+h|T} = \frac{1}{T} \sum_{t=1}^T y_t$$

- Want something in between these methods.
- Most recent data should have more weight.

Simple Exponential Smoothing

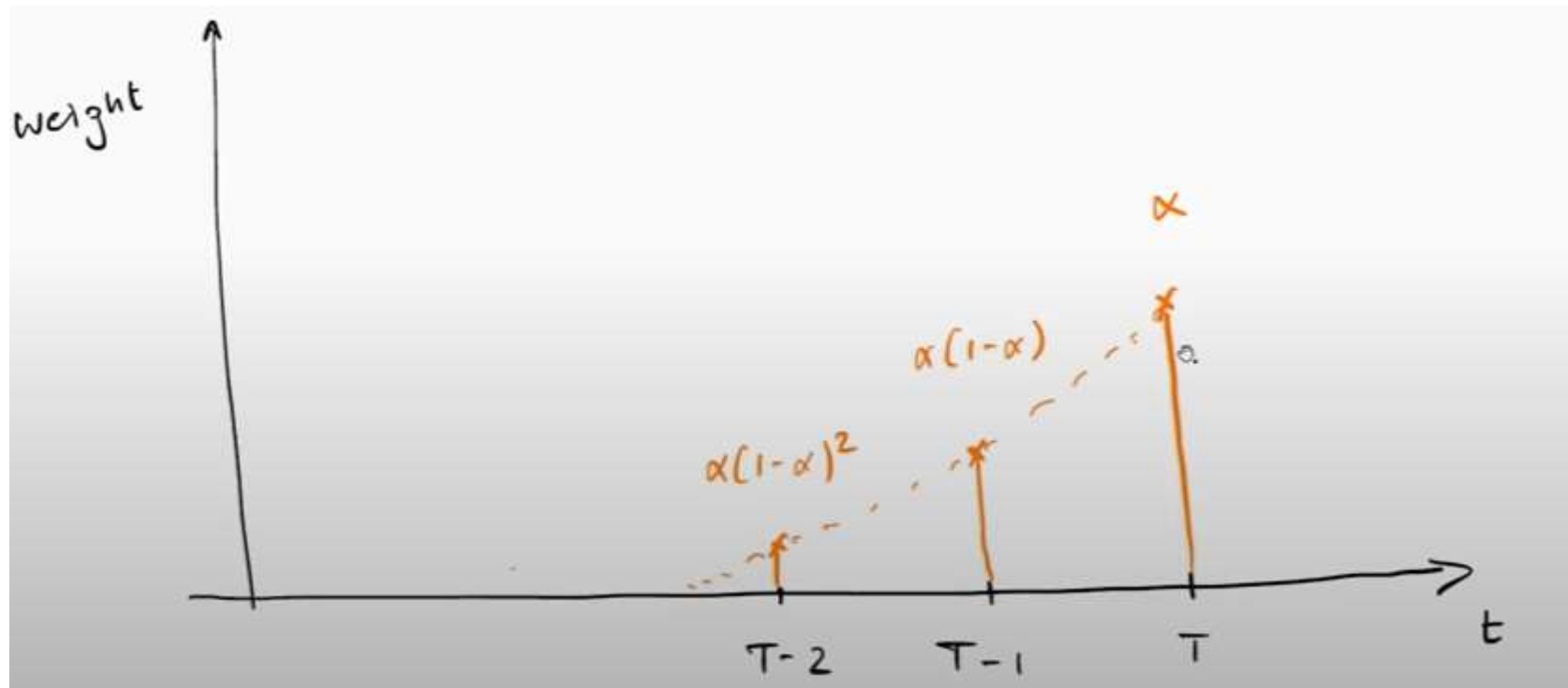
Forecast equation

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \cdots,$$

where $0 \leq \alpha \leq 1$.

Weights assigned to observations for:				
Observation	$\alpha = 0.2$	$\alpha = 0.4$	$\alpha = 0.6$	$\alpha = 0.8$
y_T	0.2	0.4	0.6	0.8
y_{T-1}	0.16	0.24	0.24	0.16
y_{T-2}	0.128	0.144	0.096	0.032
y_{T-3}	0.1024	0.0864	0.0384	0.0064
y_{T-4}	$(0.2)(0.8)^4$	$(0.4)(0.6)^4$	$(0.6)(0.4)^4$	$(0.8)(0.2)^4$
y_{T-5}	$(0.2)(0.8)^5$	$(0.4)(0.6)^5$	$(0.6)(0.4)^5$	$(0.8)(0.2)^5$

Simple Exponential Smoothing



Simple Exponential Smoothing

Weighted average form

The forecast at time $T + 1$ is equal to a weighted average between the most recent observation y_T and the previous forecast $\hat{y}_{T|T-1}$:

$$\hat{y}_{T+1|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1},$$

where $0 \leq \alpha \leq 1$ is the smoothing parameter. Similarly, we can write the fitted values as

$$\hat{y}_{t+1|t} = \alpha y_t + (1 - \alpha) \hat{y}_{t|t-1},$$

for $t = 1, \dots, T$. (Recall that fitted values are simply one-step forecasts of the training data.)

The process has to start somewhere, so we let the first fitted value at time 1 be denoted by ℓ_0 (which we will have to estimate). Then

$$\hat{y}_{2|1} = \alpha y_1 + (1 - \alpha) \ell_0$$

$$\hat{y}_{3|2} = \alpha y_2 + (1 - \alpha) \hat{y}_{2|1}$$

$$\hat{y}_{4|3} = \alpha y_3 + (1 - \alpha) \hat{y}_{3|2}$$

$$\vdots$$

$$\hat{y}_{T|T-1} = \alpha y_{T-1} + (1 - \alpha) \hat{y}_{T-1|T-2}$$

$$\hat{y}_{T+1|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1}.$$

Simple Exponential Smoothing

Substituting each equation into the following equation, we obtain

$$\begin{aligned}\hat{y}_{3|2} &= \alpha y_2 + (1 - \alpha) [\alpha y_1 + (1 - \alpha) \ell_0] \\ &= \alpha y_2 + \alpha(1 - \alpha) y_1 + (1 - \alpha)^2 \ell_0 \\ \hat{y}_{4|3} &= \alpha y_3 + (1 - \alpha) [\alpha y_2 + \alpha(1 - \alpha) y_1 + (1 - \alpha)^2 \ell_0] \\ &= \alpha y_3 + \alpha(1 - \alpha) y_2 + \alpha(1 - \alpha)^2 y_1 + (1 - \alpha)^3 \ell_0 \\ &\vdots \\ \hat{y}_{T+1|T} &= \sum_{j=0}^{T-1} \alpha(1 - \alpha)^j y_{T-j} + (1 - \alpha)^T \ell_0.\end{aligned}$$

The last term becomes tiny for large T . So, the weighted average form leads to the same forecast Equation (8.1).

Component form

An alternative representation is the component form. For simple exponential smoothing, the only component included is the level, ℓ_t .

Forecast equation	$\hat{y}_{t+h t} = \ell_t$
Smoothing equation	$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1},$

where ℓ_t is the level (or the smoothed value) of the series at time t . Setting $h=1$ gives the fitted values, while setting $t=T$ gives the true forecasts beyond the training data.

Optimizing smoothing parameters

- Need to choose best values for α and ℓ_0 .
- Similarly to regression, choose optimal parameters by minimising SSE:

$$\text{SSE} = \sum_{t=1}^T (y_t - \hat{y}_{t|t-1})^2.$$

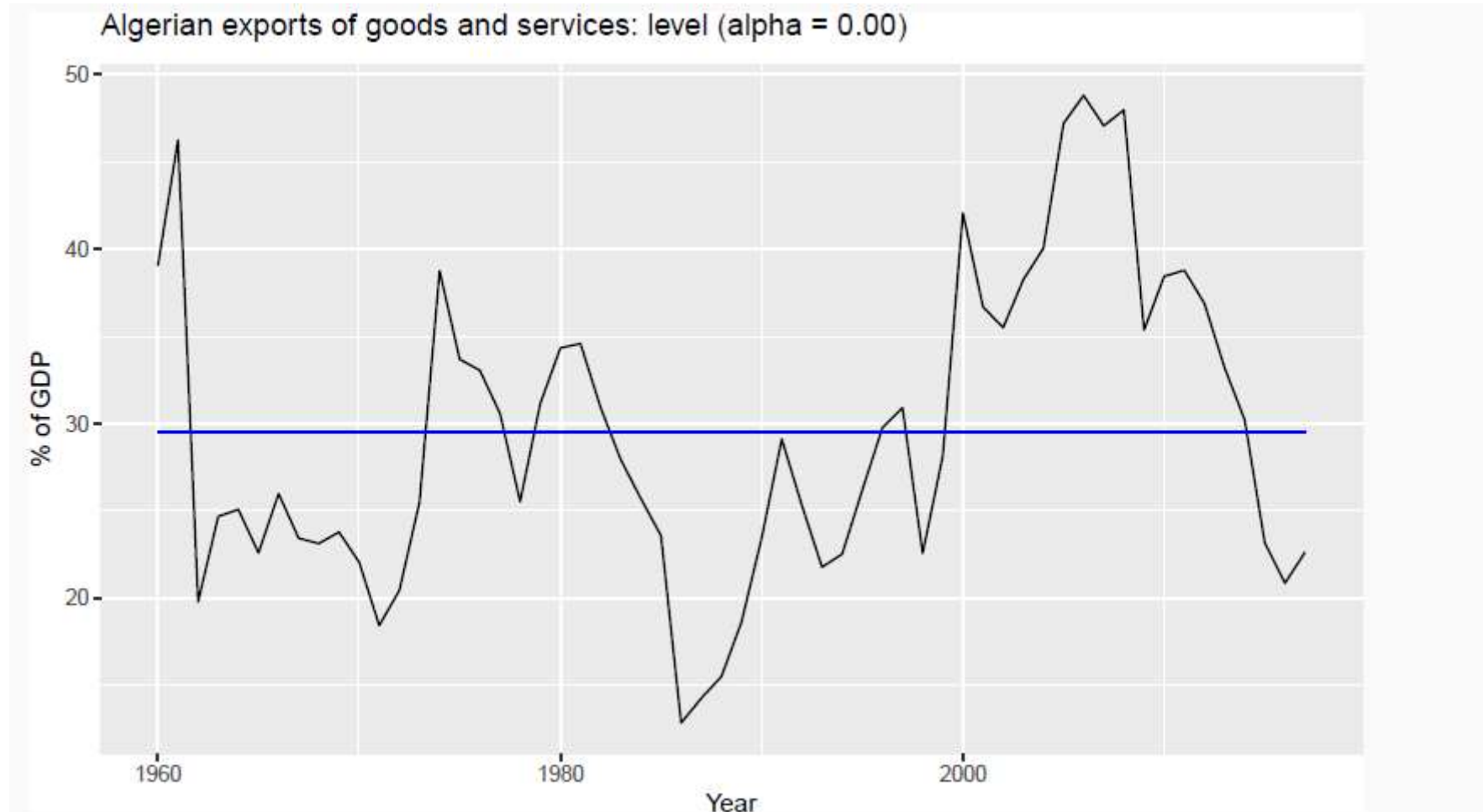
- Unlike regression there is no closed form solution — use numerical optimization.
- For Algerian Exports example:
 - ▶ $\hat{\alpha} = 0.8400$
 - ▶ $\hat{\ell}_0 = 39.54$

Example: Algerian exports

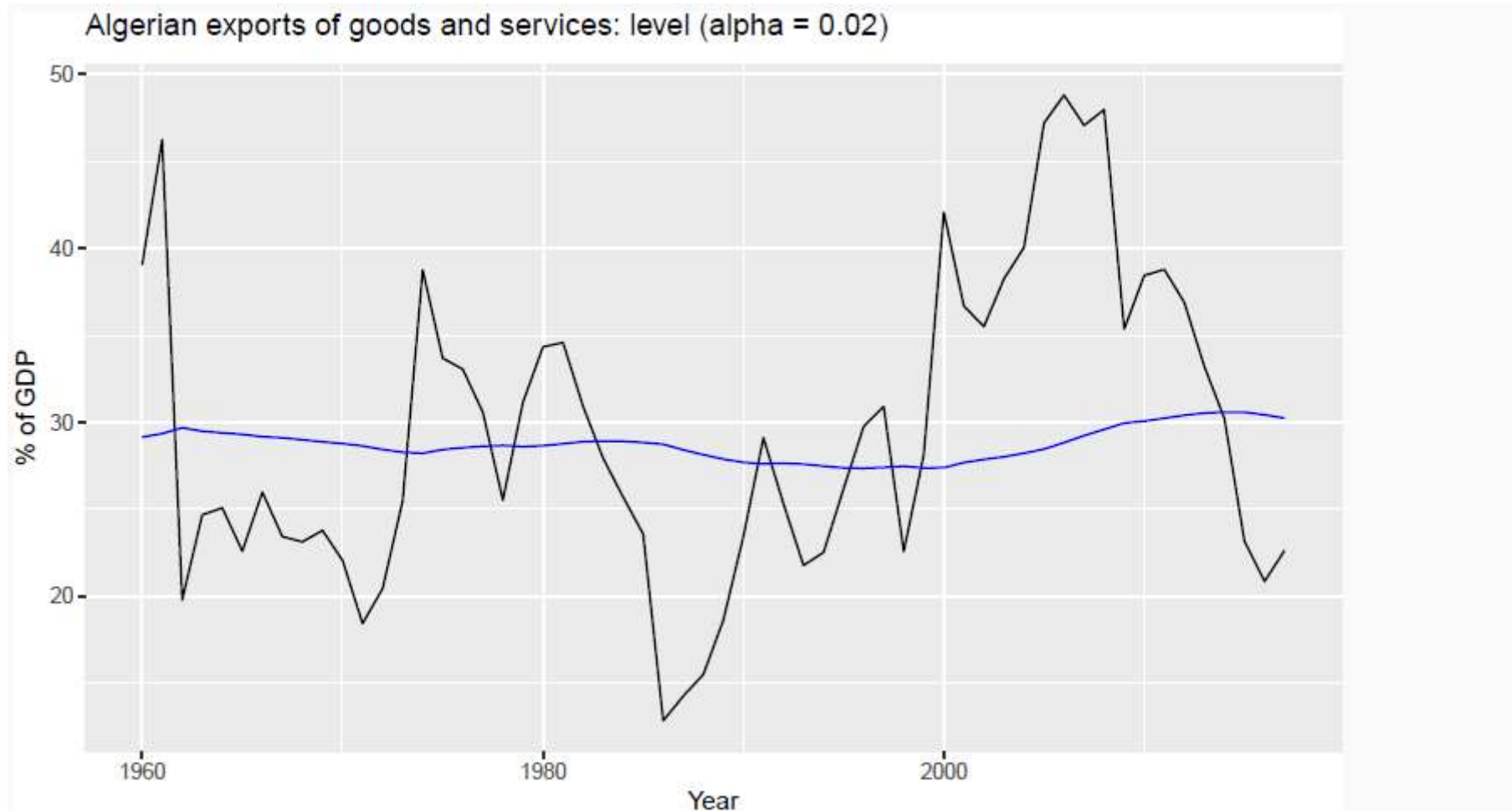
<https://otexts.com/fpp3/ses.html>

Year	Time	Observation	Level	Forecast
	t	y_t	ℓ_t	$\hat{y}_{t t-1}$
1959	0		39.54	
1960	1	39.04	39.12	39.54
1961	2	46.24	45.10	39.12
1962	3	19.79	23.84	45.10
1963	4	24.68	24.55	23.84
1964	5	25.08	25.00	24.55
1965	6	22.60	22.99	25.00
1966	7	25.99	25.51	22.99
1967	8	23.43	23.77	25.51
	$\vdots$	$\vdots$	$\vdots$	$\vdots$
2014	55	30.22	30.80	33.85
2015	56	23.17	24.39	30.80
2016	57	20.86	21.43	24.39
2017	58	22.64	22.44	21.43
	h			$\hat{y}_{T+h T}$
2018	1			22.44
2019	2			22.44
2020	3			22.44
2021	4			22.44
2022	5			22.44

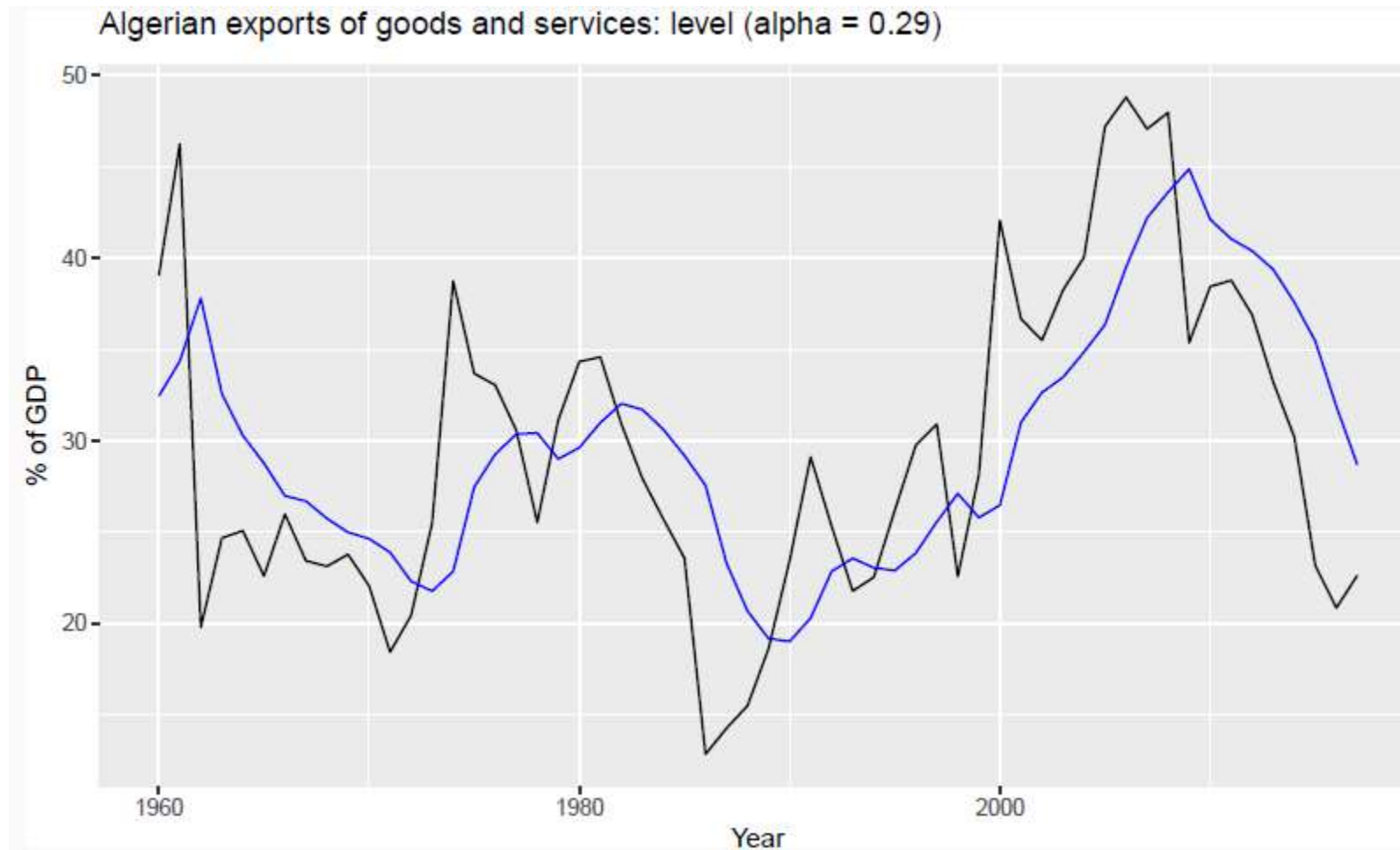
Simple Exponential Smoothing



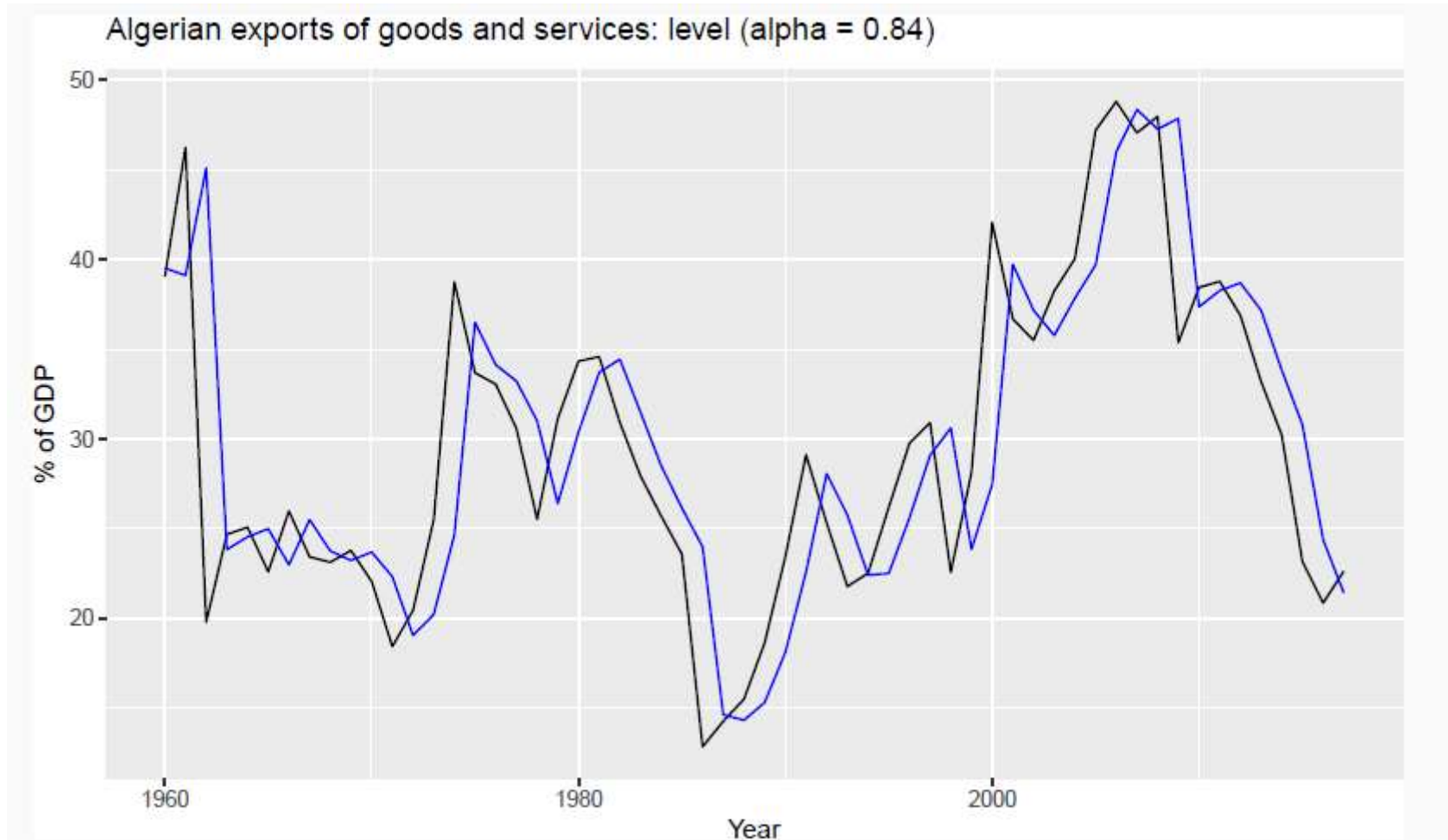
Simple Exponential Smoothing



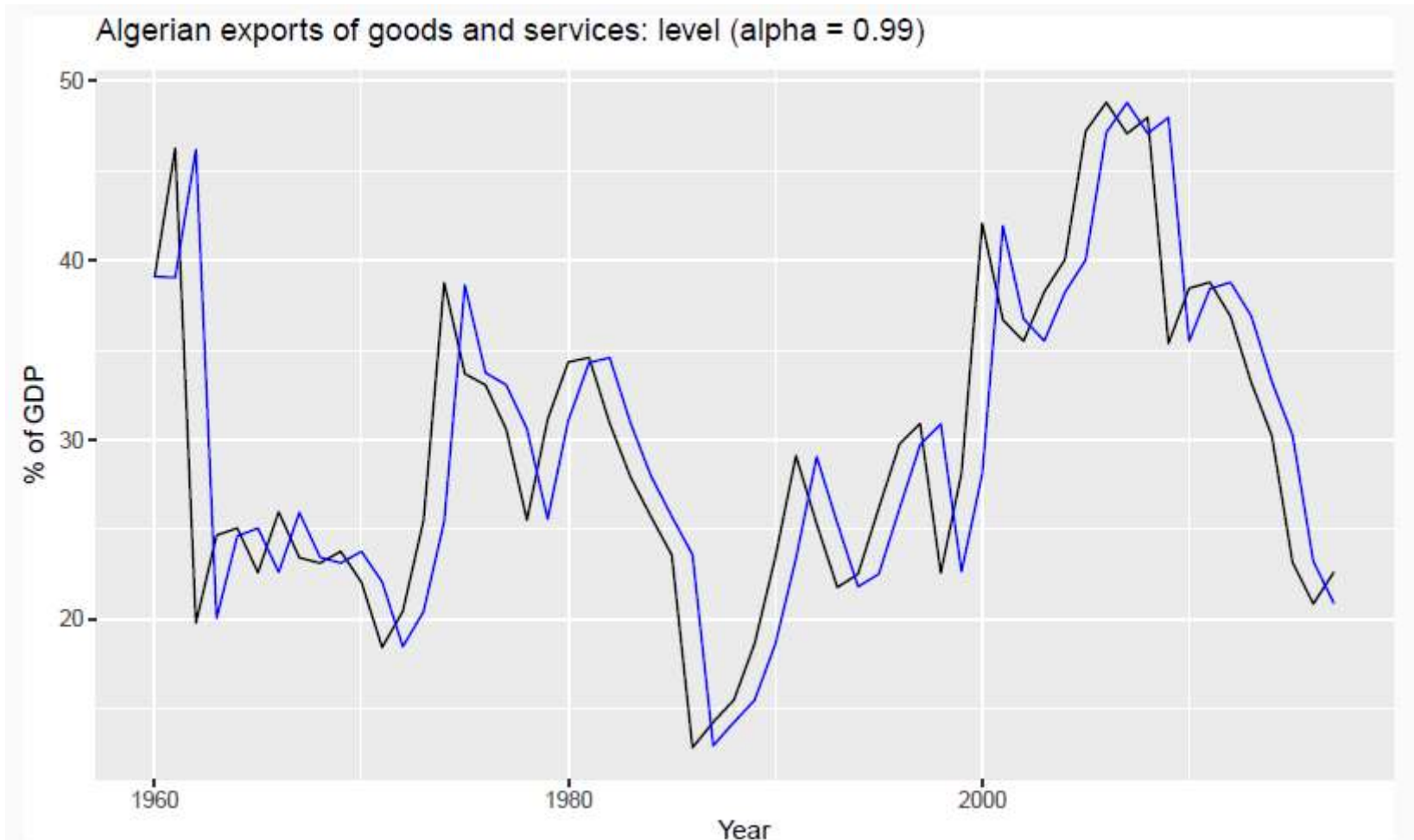
Simple Exponential Smoothing



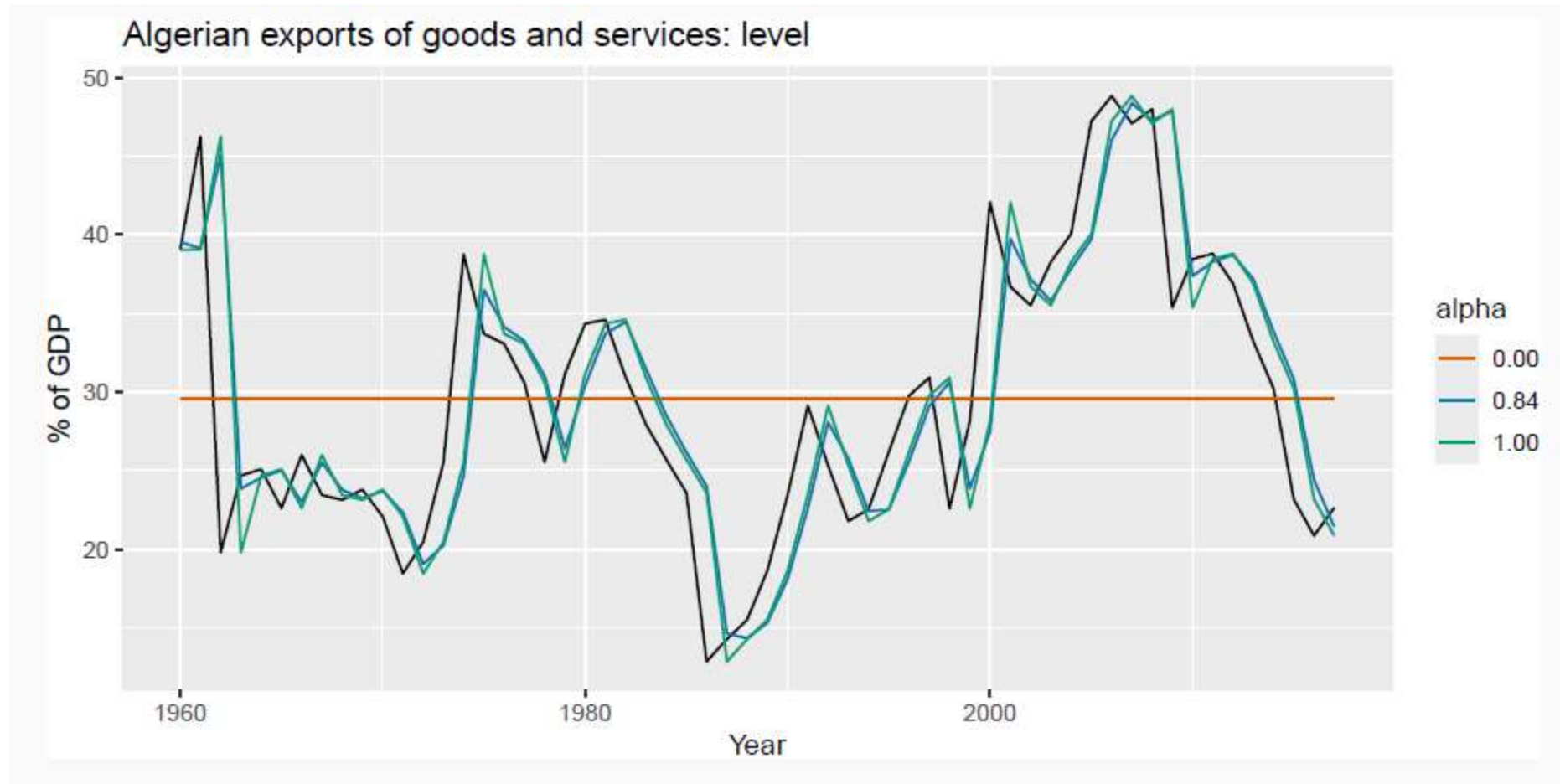
Simple Exponential Smoothing



Simple Exponential Smoothing



Simple Exponential Smoothing



Methods with trend - Holt's linear trend

Component form

Forecast

$$\hat{y}_{t+h|t} = \ell_t + hb_t$$

Level

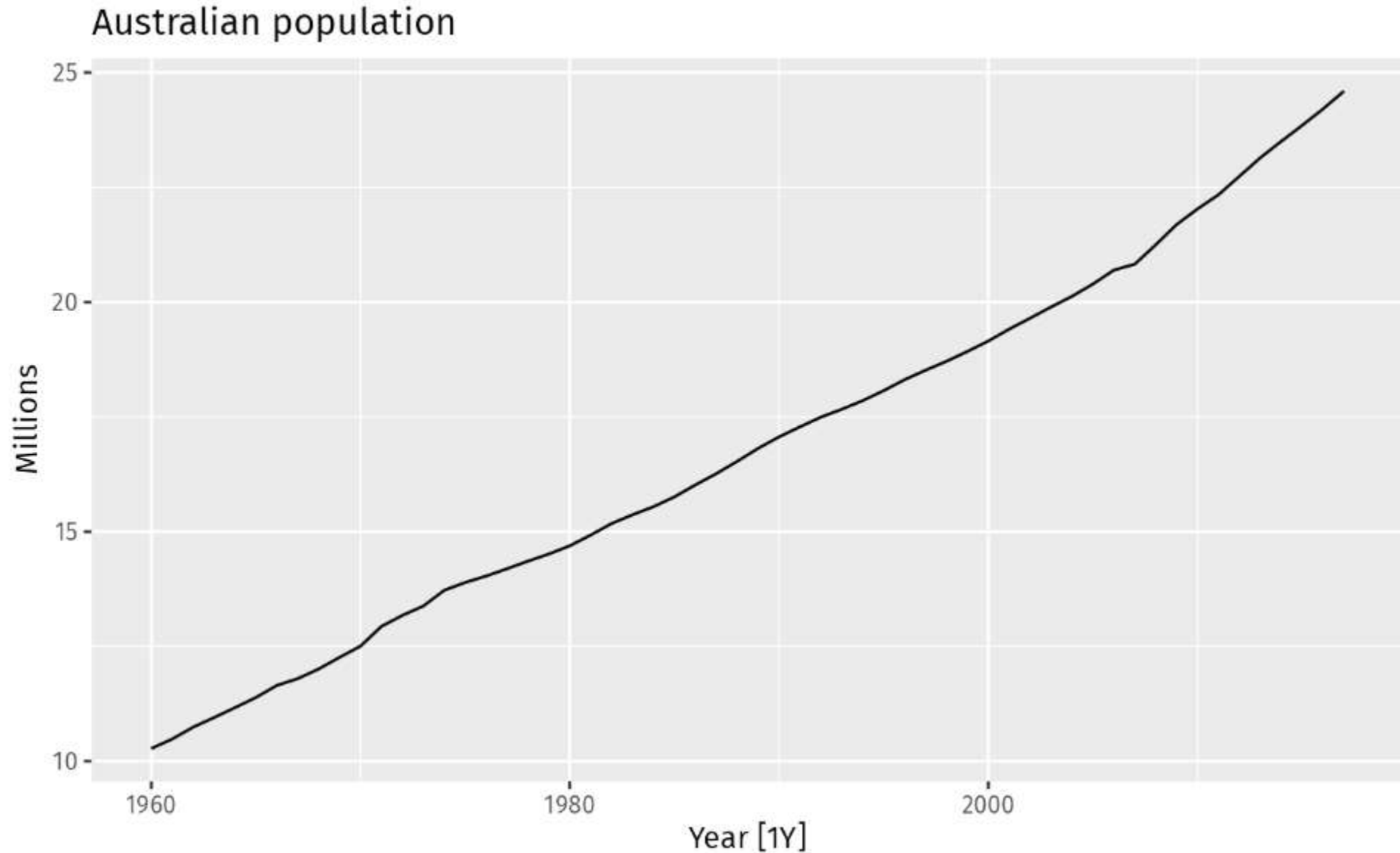
$$\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

Trend

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1},$$

- Two smoothing parameters α and β^* ($0 \leq \alpha, \beta^* \leq 1$).
- ℓ_t level: weighted average between y_t and one-step ahead forecast for time t , ($\ell_{t-1} + b_{t-1} = \hat{y}_{t|t-1}$)
- b_t slope: weighted average of $(\ell_t - \ell_{t-1})$ and b_{t-1} , current and previous estimate of slope.
- Choose $\alpha, \beta^*, \ell_0, b_0$ to minimise SSE.

Example: Australian population



Example: Australian population

Year	Time	Observation	Level	Slope	Forecast
	t	y_t	ℓ_t		$\hat{y}_{t+1 t}$
1959	0		10.05	0.22	
1960	1	10.28	10.28	0.22	10.28
1961	2	10.48	10.48	0.22	10.50
1962	3	10.74	10.74	0.23	10.70
1963	4	10.95	10.95	0.22	10.97
1964	5	11.17	11.17	0.22	11.17
1965	6	11.39	11.39	0.22	11.39
1966	7	11.65	11.65	0.23	11.61
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
2014	55	23.50	23.50	0.37	23.52
2015	56	23.85	23.85	0.36	23.87
2016	57	24.21	24.21	0.36	24.21
2017	58	24.60	24.60	0.37	24.57
	h				$\hat{y}_{T+h T}$
2018	1				24.97
2019	2				25.34
2020	3				25.71
2021	4				26.07
2022	5				26.44
2023	6				26.81
2024	7				27.18
2025	8				27.55
2026	9				27.92
2027	10				28.29

Methods with trend - Holt's linear trend

Holt and Winters extended Holt's method to capture seasonality.

Component form

$$\hat{y}_{t+h|t} = \ell_t + hb_t + s_{t+h-m(k+1)}$$

$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

- $k = \text{integer part of } (h - 1)/m$. Ensures estimates from the final year are used for forecasting.
- Parameters: $0 \leq \alpha \leq 1$, $0 \leq \beta^* \leq 1$, $0 \leq \gamma \leq 1 - \alpha$ and $m = \text{period of seasonality}$ (e.g. $m = 4$ for quarterly data).

Methods with seasonality- Holt-Winters additive method

Holt and Winters extended Holt's method to capture seasonality.

Component form

$$\hat{y}_{t+h|t} = \ell_t + hb_t + s_{t+h-m(k+1)}$$

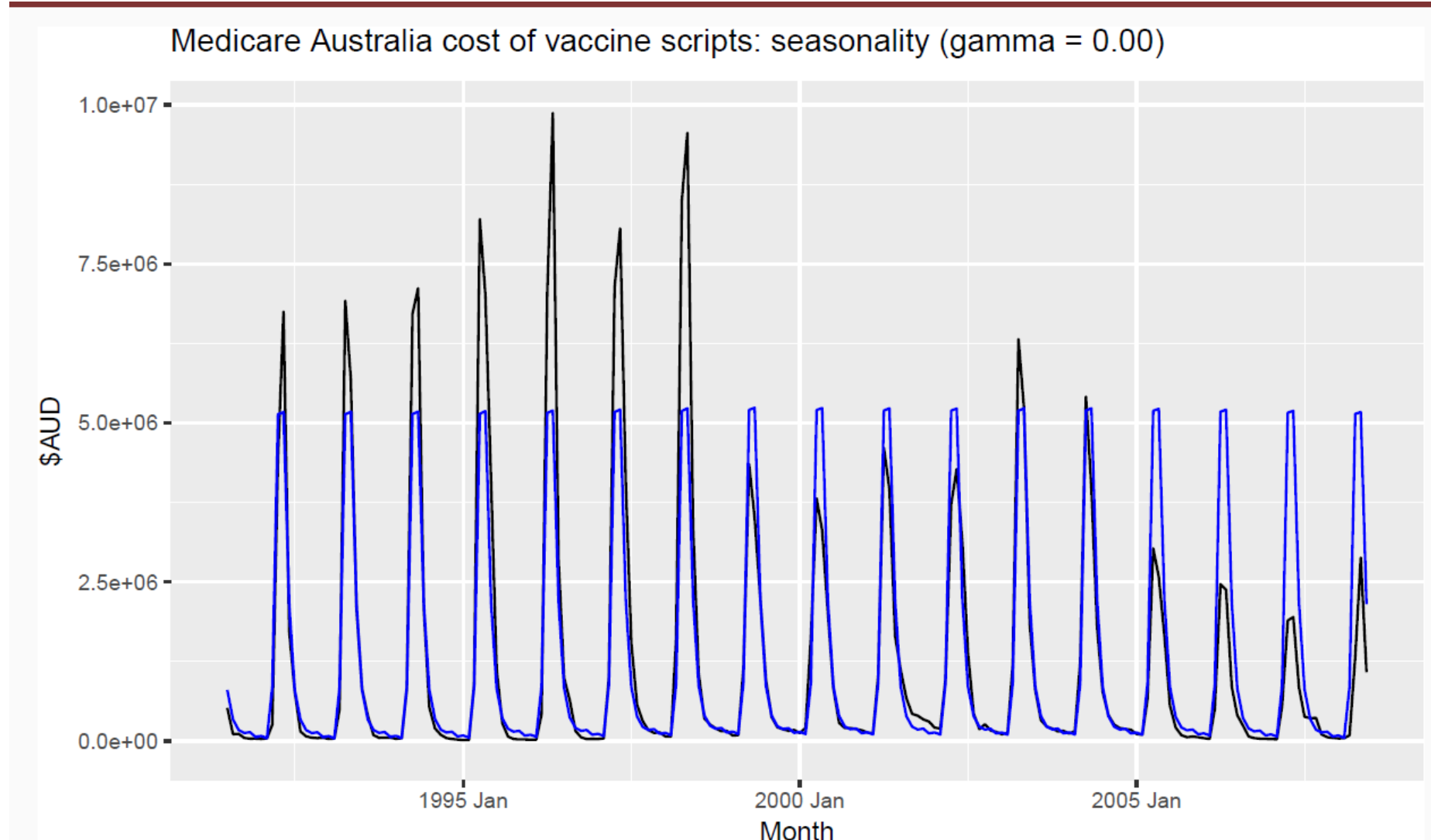
$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$$

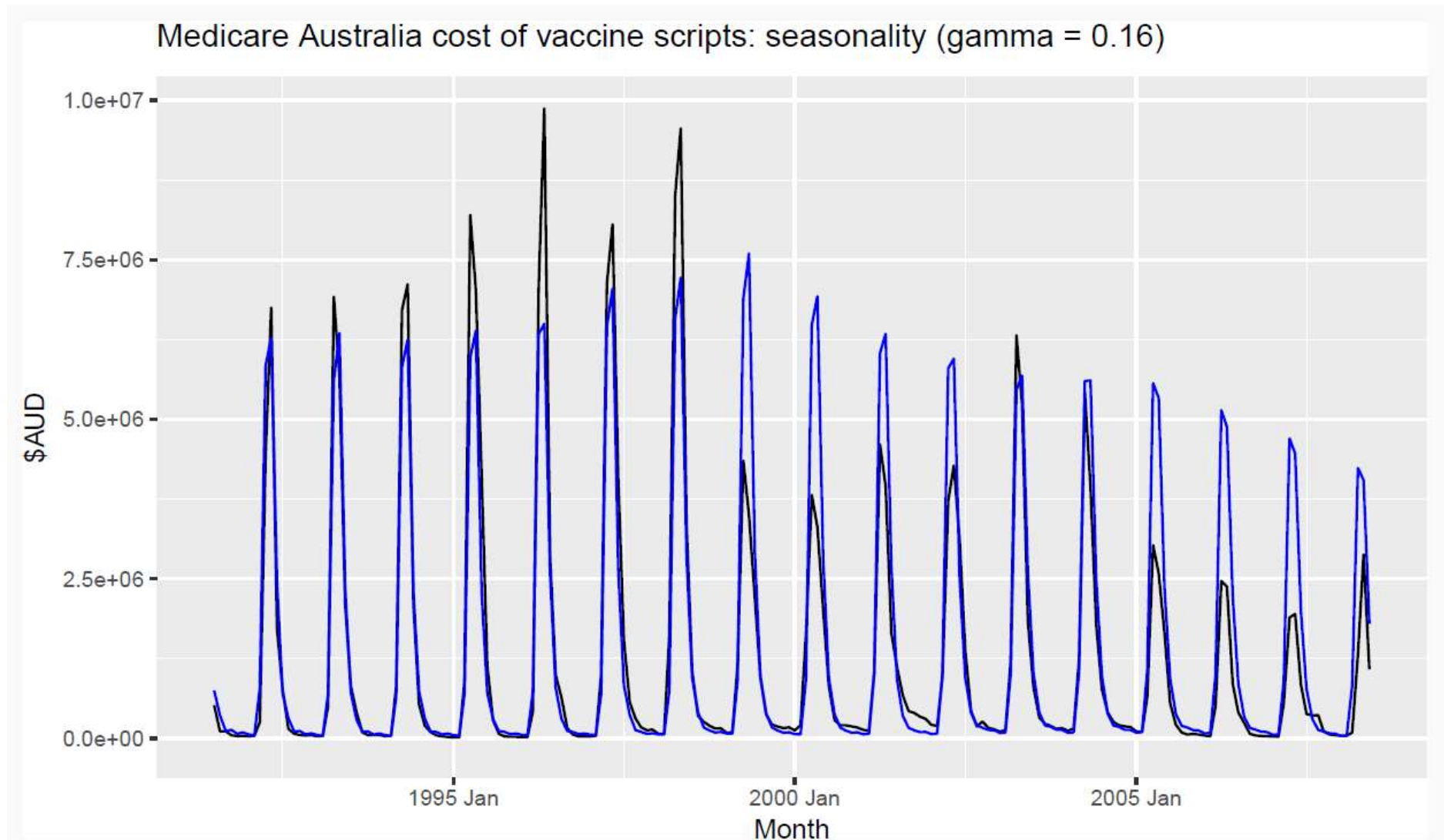
$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

- $k = \text{integer part of } (h - 1)/m$. Ensures estimates from the final year are used for forecasting.
- Parameters: $0 \leq \alpha \leq 1$, $0 \leq \beta^* \leq 1$, $0 \leq \gamma \leq 1 - \alpha$ and $m = \text{period of seasonality}$ (e.g. $m = 4$ for quarterly data).

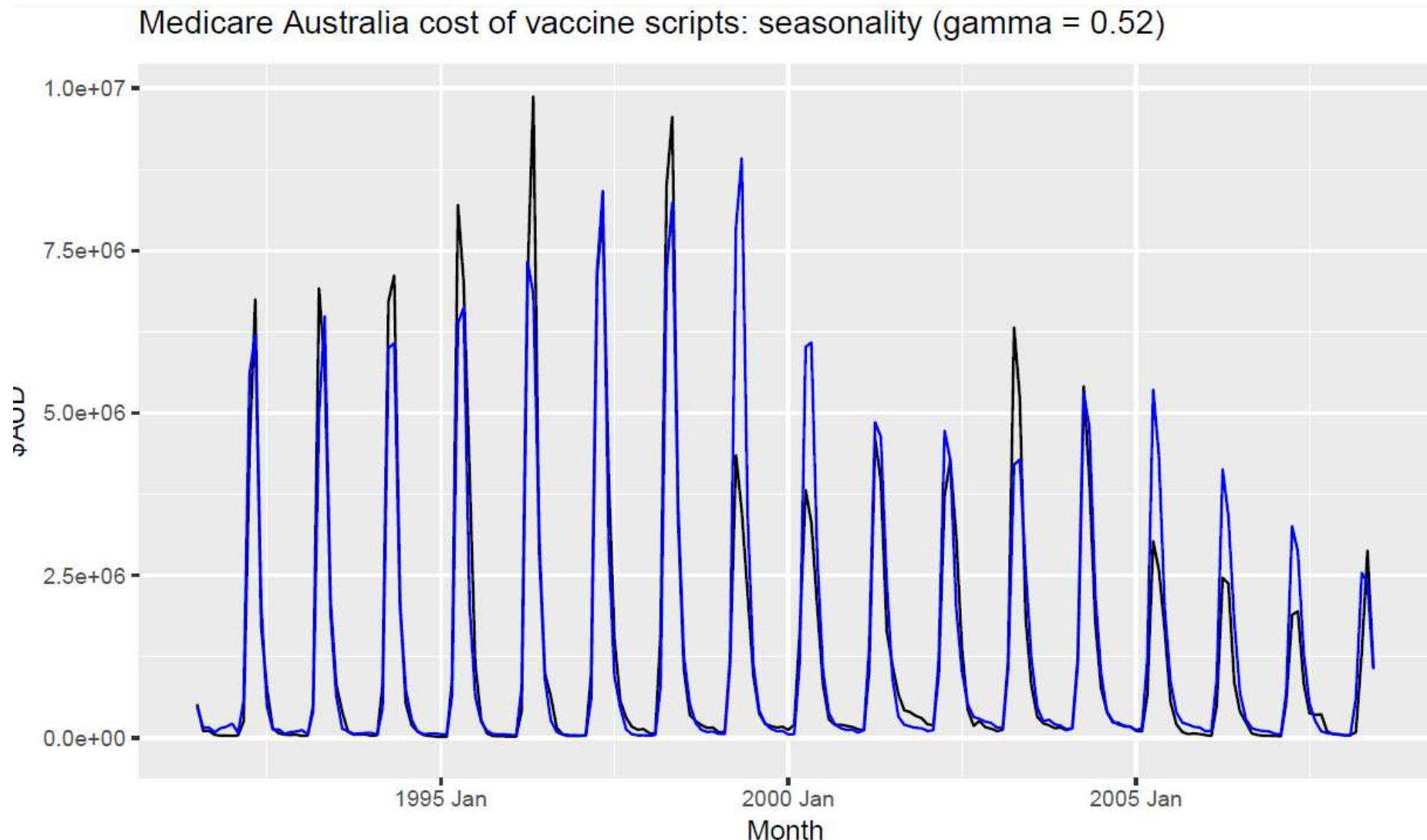
Exponential smoothing: seasonality



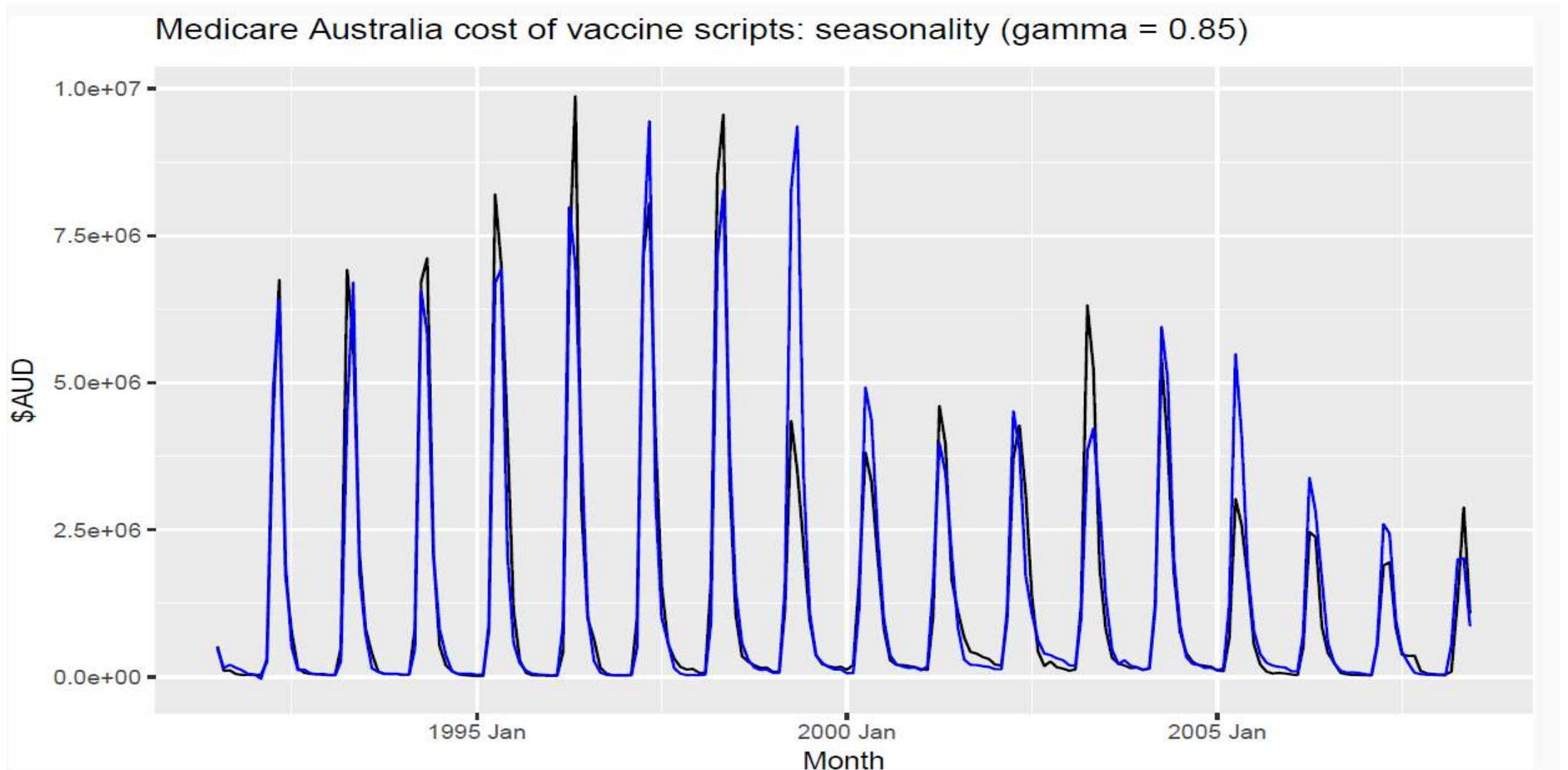
Exponential smoothing: seasonality



Exponential smoothing: seasonality



Exponential smoothing: seasonality



Methods with seasonality-Holt-Winters multiplicative method

Seasonal variations change in proportion to the level of the series.

Component form

$$\hat{y}_{t+h|t} = (\ell_t + hb_t)s_{t+h-m(k+1)}$$

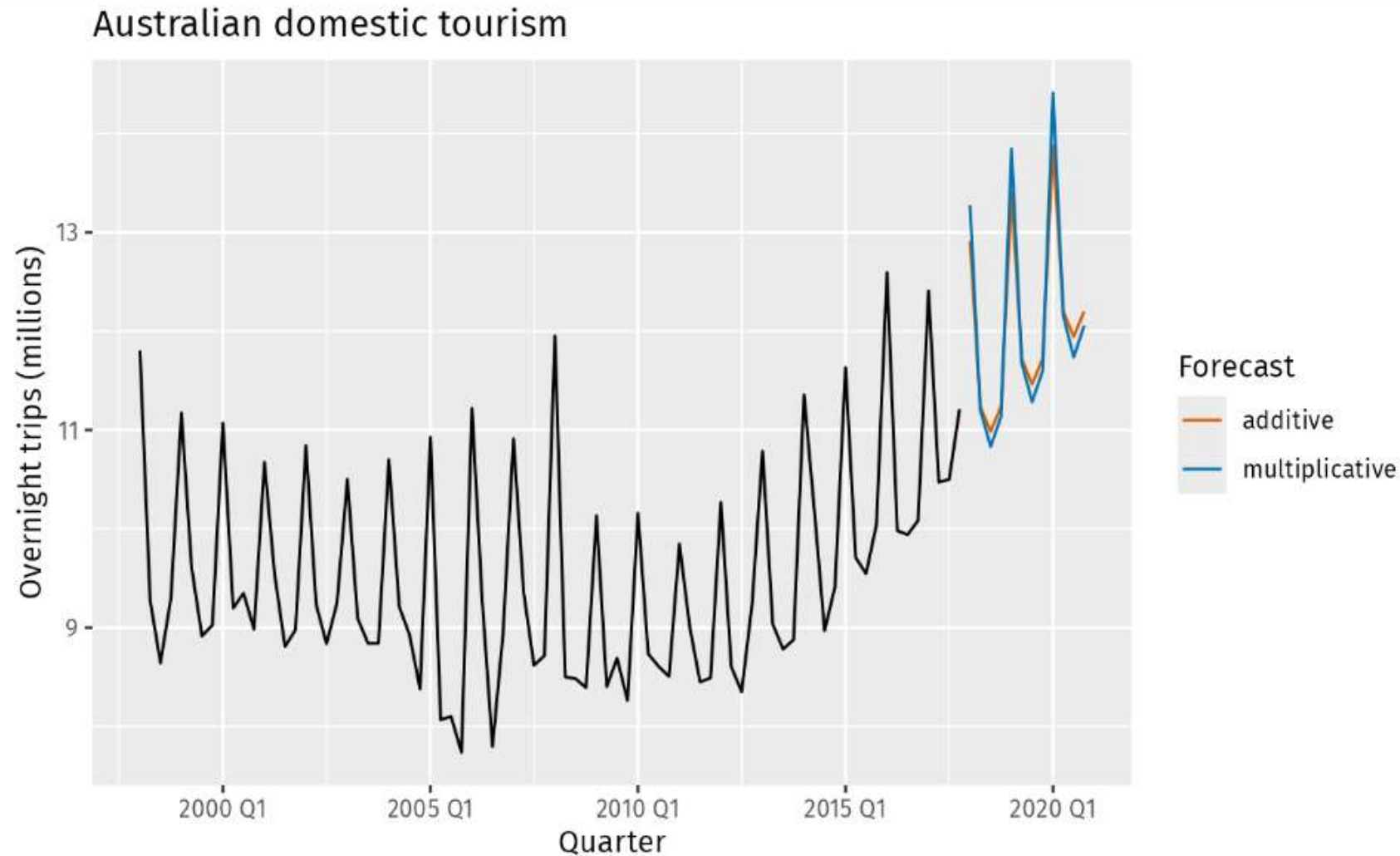
$$\ell_t = \alpha \frac{y_t}{s_{t-m}} + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$$

$$s_t = \gamma \frac{y_t}{(\ell_{t-1} + b_{t-1})} + (1 - \gamma)s_{t-m}$$

- k is integer part of $(h - 1)/m$.
- Additive method: s_t in absolute terms — within each year $\sum_i s_i \approx 0$.
- Multiplicative method: s_t in relative terms — within each year $\sum_i s_i \approx m$.

Methods with seasonality-Holt-Winters multiplicative method



Example: Domestic overnight trips in Australia

